Evidence Data Lab: Experimental Approaches to Campus Decarbonization

Course Syllabus, Revised 16.04.2026

Teaching Team Information

Lecturer: Nils Handler
Emails: nils.handler@uzh.ch
Office Hours: By appointment
Teaching Assistants: Nicolás Harrington Ruiz (nicolas.harringtonruiz@uzh.ch), Finn Ole Semrau (finnole.semrau@ifw-kiel.de)

Course Logistics

Date and Time: Weekly, Wednesday 14:00-15:45 (February-May 2025)
Location: KOL-F-123

Pre-requisites and Prior Knowledge

Basic knowledge of statistics and data analysis is a strong asset for the course, but is not required. Coding is also a useful skill for this course, particularly R, Python, or STATA.

Important Note

To accommodate the composition of different backgrounds of students in the course, it is possible that the main topics and structure of the reports will be adapted to better suit these conditions. The outline listed below will serve as the baseline for the course and is the plan for the students. Students can reasonably expect to cover the topics and produce the deliverables outlined in this syllabus.

Course Description

As organizations look to decarbonize, they have several different methods to choose from. The development of these methods requires that they are tested to ensure theoretical emissions reductions turn into actual emissions reductions. The randomized controlled trial (RCT), common in other experimental settings, remains an important tool for assessing these sustainability programs, and the goal of this course is to guide students through the process of deploying an RCT in a university setting.

The course will include an introduction to sustainability measures at UZH to familiarize students with what has already been undertaken at UZH. The main body of the course will instruct students on RCTs, namely:

1. What an RCT is, what a good one looks like, and the intuition and mathematical structures that make it an important tool.
2. How to structure ideas for developing an RCT through the current conventional approach or identify the need for an RCT through a literature review.
3. How to reflect on other RCT designs to highlight issues that could impact results and provide critiques for improving them.

Students will navigate these subjects through a series of lectures introducing these concepts and their supporting mathematical framework, including how causality, randomization, hypothesis design, and power form critical aspects for a good RCT.

During a typical lectures, students will engage with the week's topic through a lecture during the first 60 minutes of the course and then a 30-minute discussion on applying this subject to their project and other questions they might have on the work to be done as part of the course.

Outside of lectures (and during some workshop sessions in class), students will either develop their own RCT based on selected topics provided by the teaching team or identify where an RCT could take place by reviewing the literature concerning sustainability. Students should consult the list of suggested topics provided by the teaching team to choose an area for this project, though students can propose a topic for approval. Through intermediate progress update presentations and feedback, students will be expected to refine this project to a state where it could provide a guide to researchers wishing to carry out that experiment.

Objectives

Upon completion of the module, students will be able to:

Independently identify the need for or plan and evaluate RCTs for creating causal evidence fields, such as campus decarbonization.

1. The “plan” goal will be accomplished by writing a pre-analysis plan and supporting documents, such that researchers could follow those instructions to perform the RCT.
2. The “identify the need” goal will be accomplished by conducting a literature review on the topic that concisely discusses the current state of the field and where future RCTs should be conducted.

Clearly communicate scientific findings and formulate methodologically sound recommendations for action.

1. The “communicate” goal will be accomplished by writing a clear report that communicates the findings of the student’s work, such that researchers in the field could build on it clearly by executing the planned RCT or developing the RCT identified as a potential gap in the literature.
2. The “formulate methodologically sound recommendations” goal will be accomplished by practicing the identification or development of hypotheses and interventions for their chosen sustainability topic of choice based on the current state of the literature or the provided topic in sustainability.

Course Materials

The OLAT for the course will contain all slides presented in class. In addition to these basic resources, reference lists for each subject will be provided, primarily drawn from the behavioural economics literature and teaching materials from organizations working in behavioural and development economics. Additional supplementary references will also be provided.

Reference papers will be provided for each topic on the OLAT page. These case studies will provide examples of both the finished version of an experiment and how it was planned, along with references for important context. Students can request additional reference papers throughout the course.

Some of the lectures will feature external speakers, and it is planned to make the recordings of these lectures available to students, along with any materials presented during those lectures.

Evaluation and Grading

Assignments	% of Grade
Final Project	80%
In-class participation	20%

Final Project – 80% of the final grade

There are two options for the final project: the experiment development project and the literature review project. They are explained below and operate on the same timeline. Grading for the final project will occur after the final submission at the end of the course, but there will be opportunities for feedback around the intermediary presentation of progress on each student's project. Timely submission for intermediary deadlines will factor into the grade.

The project can be completed and submitted in groups of one to three students. The presentations during the course would consist of a summary of the main elements of the report and, during the intermediary phases, an outline of remaining open questions that the student(s) would like to have their peers discuss.

Experiment Development Project

The Experiment Development Project consists of the submission of the pre-analysis plan at the end of the course. A pre-analysis plan is a document submitted during the pre-registration of a study. It outlines the justifications and plans for the execution of a particular study, as well as planned methods. These documents are part of a process that has developed in several sciences to help combat the reproducibility crisis and the prevalence of p-hacking. As there can be a high variance in the structure of pre-analysis plans, the format that will be followed will be from examples provided by the teaching team, with a focus on those found on the American Economic Association (AEA) Social Science Registry (example here). More details will be included during the course.

Literature Review Project

The literature review project consists of students reviewing a topic to identify gaps in the literature concerning the application of RCTs in sustainability. Students should, starting from the provided literature, expand their search to identify open questions posed in the research of their topic that have yet to be extensively covered. The purpose of this project is to enable students to practice identifying where to conduct an experiment when entering a field or confronted with a problem requiring solutions from the academic literature, such as sustainability on university campuses.

In-class Participation – 20% of the final grade

The in-class participation component is primarily composed of participating in group discussions during the presentations of the case studies and pre-analysis plans and providing feedback for their peers. General attendance will also be considered, as consistent unannounced absences would hinder group discussions.

Draft Schedule

Lecture(s)/Date	Item	Description
Lectures 1-3	RCT and literature review introduction	Introduction to the main elements of RCT development During this time, students will take what they are learning in class, and the options presented to them and select a topic and relevant case study by lectures 3-4.
Finished by lecture 3	Topic confirmation	Students will briefly confirm their selected topics (and groups) before one of the lectures in the introduction to RCTs section.
Lectures 4-7	Experimental design and review, in-depth	Building on the concepts of the RCT introduction, we will continue discussing the mathematical frameworks of RCTs in more depth, including causality, hypothesis design, power calculations, and randomization. Students will begin working on their project topic outside of the lecture time (with some workshops for small feedback).
Lecture 8 or 9	Intermediate update presentation	Students will present a summary of the current state of the project, along with open questions they have, so that the group can discuss them and provide feedback. This presentation should have clearly outlined hypotheses and considered basic questions of data collection.
Lectures 10-14	RCT analysis and additional context	A continuation of the previous RCT in-depth module, focusing on data collection, analysis, and practical issues. Students will continue to work on their projects for the final submission and presentation.
Lecture on 27.05.2026	Final presentation and submission	Students will present their final report. Peer feedback will follow, and graded feedback will follow from the submission of the full report package.

Course Modules & Timeline

The course is structured into two main parts: **Design** and **Methods**. The first half (**Design**) focuses on the conceptual foundations of experimental design, while the second half (**Methods**) provides a deeper dive into the analytical tools for causal inference.

Week	Date	Module	Theme	Details
1	Feb 18	Design	Welcome & Intro to RCTs	Orientation on the course, its topics, and the projects to be submitted
2	Feb 25	Design	Introduction to PAPs + UZH Sustainability	Why PAPs are important and how to structure them + UZH Sustainability guest
3	Mar 4	Design	Introduction to Literature Reviews	Skills on how to identify where to start building your next RCT
4	Mar 11	Skipped	No Lecture	Potentially some out-of-class reading and other materials
5	Mar 18	Skipped	No Lecture	Potentially some out-of-class reading and other materials
6	Mar 25	Design	Hypothesis Design & Power Calculations	One and two-sided tests, multiple testing, power calculations
7	Apr 1	Design	Causality	Understanding causality vs. correlation, confounding, DAGs
8	Apr 8	Break	No Lecture	Continue Project Work
9	Apr 15	Design	Randomization	Stratification, clustering

Week	Date	Module	Theme	Details
10	Apr 22	Methods	**PAP First Draft Due** & Peer Review Workshop	Presenting initial project ideas and providing constructive feedback.
11	Apr 29	Methods	Analysis of Results	Compliers, defiers, LATE, and other methods
12	May 6	Methods	Practical Issues in research	Ethics, implementation, recruitment, and dealing with attrition and imperfect compliance
13	May 13	Methods	**Bonus lecture**	**To be determined by student preferences?**
14	May 20	Bonus Lecture	TBD	To be decided based on student interests
15	May 27	Submit	**Final Project Presentations**	Students present their experimental design, analysis, and findings to the class.

The schedule above will have some flexibility as to the exact presentation dates. More details for presentations and the dates of guest speakers will be provided as they are confirmed. The sequence in which subjects are shown should be preserved, but the exact dates may change as guest speakers are confirmed.